$Var(S(\theta; y))$

Fisher Information

- Fisher information is the variance of the score function
- It tells us how much information Y carries about the parameters of the distribution that models Y
- The asymptotic variance of the MLE is the inverse of the Fisher information

Fisher information

Suppose that $Y_1, Y_2, \ldots, Y_n$ are i.i.d. with pdf $f(\boldsymbol{y}; \theta)$. Subject to regularity conditions on $f(\boldsymbol{y}; \theta)$, I_θ is known as the **Fisher information** about θ in the observations.

$$S(\theta; y) = \frac{\partial \ell}{\partial \theta}$$

$$I_\theta = \text{var}\big(S(\theta; \boldsymbol{Y})\big) = E\left[\left(\frac{\partial \ell}{\partial \theta}\right)^2\right] = E\left(S^2(\theta, Y)\right)$$

$$= E\left[S^2(\theta; Y)\right] - E^2\left[S(\theta; Y)\right]$$

$$E^2\left[S(\theta; Y)\right] = 0$$

$$I(\theta_1, \theta_2) \quad \begin{pmatrix} E\left(\frac{\partial \ell}{\partial \theta_1 \partial \theta_1}\right) & E\left(\frac{\partial \ell}{\partial \theta_1 \partial \theta_2}\right) \\ E\,\frac{\partial \ell}{\partial \theta_2 \partial \theta_1} & E\left(\frac{\partial \ell}{\partial \theta_2 \partial \theta_2}\right) \end{pmatrix}$$

If $\theta = (\theta_1, \theta_2, \ldots, \theta_k)^T$, then I_θ is the Fisher information matrix, with dimensions $k \times k$. The ijth element of I_θ is given by

$$[I_\theta]_{ij} = E\left[\frac{\partial \ell}{\partial \theta_i} \frac{\partial \ell}{\partial \theta_j}\right].$$

Precise statement of the regularity conditions is somewhat technical. But, in broad terms, we require that:

- The probability density f is sufficiently many times continuously differentiable
- The support of Y does not depend on θ

These are needed so that we can exchange the order of integration (with respect to $\boldsymbol{y}$) and differentiation (with respect to θ). That is, we can perform operations like the following:

$$\frac{\partial^2}{\partial\theta^2} \int \boxed{f(\theta;\boldsymbol{y})}\, d\boldsymbol{y} = \int \frac{\partial^2}{\partial\theta^2} f(\theta;\boldsymbol{y})\, d\boldsymbol{y}$$

Alternative form

Under the same regularity conditions as for the Cramér-Rao inequality:

$$I_\theta = -E\left[\frac{\partial^2 \ell}{\partial \theta^2}\right].$$

① ②

$$Var(S(\theta; Y)) = E\left[\left(\frac{\partial \ell}{\partial \theta}\right)^2\right]$$

In the case of $\theta = (\theta_1, \theta_2, \ldots, \theta_K)^T$,

$$[I_\theta]_{ij} = -E\left[\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_j}\right].$$

Proof of alternative form

$$E\left[\frac{\partial^2 \ell}{\partial \theta^2}\right. = \frac{\partial}{\partial \theta}\left(\frac{\partial \ell}{\partial \theta}\right)$$

$$= \frac{\partial}{\partial \theta}\left(\frac{\partial \log L}{\partial \theta}\right)$$

$$= \frac{\partial}{\partial \theta}\left(\underbrace{\frac{1}{L}}_{g}\,\frac{\partial L}{\partial \theta}\right) \to f$$

$$\frac{\partial}{\partial \theta}\left(\frac{\overset{=: f}{\dfrac{\partial L}{\partial \theta}}}{L}\right)$$

quotient rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

In our case, let $f = \frac{\partial \ell}{\partial \theta}$ and $g = L$.

Then $f' = \frac{\partial^2 \ell}{\partial \theta^2}$ and $g' = \frac{\partial L}{\partial \theta}$.

$$= \frac{\left(\frac{\partial^2}{\partial \theta^2} L\right) L - \left(\frac{\partial L}{\partial \theta}\right)\left(\frac{\partial L}{\partial \theta}\right)}{L^2}$$

$$= \frac{\frac{\partial^2 L}{\partial \theta^2}}{L} - \left[\boxed{\frac{\left(\frac{\partial L}{\partial \theta}\right)}{L}}\right]^2$$

$$\frac{\partial \log L}{\partial \theta} = \boxed{\frac{1}{L}\left(\frac{\partial L}{\partial \theta}\right)}$$

$$= \frac{1}{L}\frac{\partial^2 L}{\partial \theta^2} - \left[\frac{\partial \log L}{\partial \theta}\right]^2 \to S^2(\theta; y)$$

Proof of alternative form

$$E\left[-\frac{\partial^2 \ell}{\partial\theta^2}\right] = E\left[-\frac{1}{L}\frac{\partial^2 L}{\partial\theta^2} + \left(\frac{\partial \log L}{\partial\theta}\right)^2\right]$$

$$= -\boxed{E\left[\frac{1}{L}\frac{\partial^2 L}{\partial\theta^2}\right]} + \underline{E\left[\left(\frac{\partial \log L}{\partial\theta}\right)^2\right]} = I_\theta$$

$$E\left[\frac{1}{L}\frac{\partial^2 L}{\partial\theta^2}\right] = \int_{-\infty}^{\infty} \frac{1}{\cancel{L}}\frac{\partial^2 L}{\partial\theta^2}\cancel{L}\,dy$$

$$= \int_{-\infty}^{\infty} \frac{\partial^2 L}{\partial\theta^2}\,dy$$

$$= \frac{\partial^2}{\partial\theta^2}\underbrace{\int_{-\infty}^{\infty} L\,dy}_{1}$$

under regularity conditions

as we are integrating a density over its support

$$= \frac{\partial^2}{\partial\theta^2} 1$$

$$= 0$$

$$\therefore\ E\left[-\frac{\partial^2 \ell}{\partial\theta^2}\right] = E\left[\left(\frac{\partial \ell}{\partial\theta}\right)^2\right] = E\left[S(\theta;y)^2\right] = Var(S(\theta;y)) = I_\theta$$

Cramér-Rao inequality

Suppose that $Y_1, Y_2, \ldots, Y_n$ are i.i.d. with pdf $f(\boldsymbol{y};\theta)$. Subject to regularity conditions on $f(\boldsymbol{y};\theta)$, we have that for any unbiased estimator $\tilde{\theta}$ for θ,

$$var(\tilde{\theta}) \geq I_\theta^{-1}$$

where

$$I_\theta = E\left[\left(\frac{\partial \ell}{\partial \theta}\right)^2\right].$$

Example 5.7

Suppose $y_1, y_2, \ldots, y_n$ are $i.i.d.\, Po(\lambda)$ observations.
Find the Fisher information about λ.

$$\ell(\lambda; y) = -n\lambda + \sum_{i=1}^{n} y_i \log\lambda + \log \prod_{i=1}^{n}\left(\frac{1}{y_i!}\right)$$

$$S(\lambda; y) = \frac{\partial \ell}{\partial \lambda} = -n + \frac{\sum y_i}{\lambda}$$

$$\frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{1}{\lambda^2}\left(\sum y_i\right)$$

$$I_\lambda = E\left(-\frac{\partial^2 \ell}{\partial \lambda^2}\right) = E\left[\frac{1}{\lambda^2}\sum y_i\right]$$

$$= \frac{1}{\lambda^2} E\left[\sum y_i\right]$$

$$= \frac{1}{\lambda^2}\, n\, E(y_i) = \lambda$$

$$= \frac{n}{\lambda}$$

Example 5.8

Suppose $y_1, y_2, \ldots, y_n$ are $i.i.d.\, N(\mu, \sigma^2)$ observations with σ^2 known. Find the Fisher information about μ.

$$l(u; y) = -\frac{n}{2}\log(2\pi 6^2) - \frac{1}{26^2}\Sigma(y_i - u)^2$$

$$\frac{\partial l}{\partial u} = \frac{1}{6^2}\Sigma(y_i - u)$$

$$\frac{\partial^2 l}{\partial u^2} = \frac{-n}{6^2}$$

$$I_u = E\left(\frac{n}{6^2}\right) = \frac{n}{6^2}$$

$$I_{u,6^2} = \begin{bmatrix} \frac{-n}{6^2} & \boxed{1} \\ \boxed{2} & \boxed{3} \end{bmatrix}$$

Example 5.9

Suppose $y_1, y_2, \dots, y_n$ are $i.i.d.\, N(\mu, \sigma^2)$ observations **where both μ and σ^2 are unknown**. Find the Fisher information matrix.

$$\ell(u, 6^2; y) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(6^2) - \frac{1}{26}\Sigma(y_i - u)^2$$

$$\frac{\partial^2 \ell}{\partial u \partial 6^2} = \frac{\partial \frac{1}{6^2}\Sigma(y_i - u)}{\partial 6^2} = -\frac{1}{64}\Sigma(y_i - u)$$

$$\parallel$$

$$\frac{\partial \ell}{\partial 6^2 \partial u}$$

$$\frac{\partial^2 \ell}{\partial 64} = \frac{n}{264} - \frac{1}{66}\Sigma(y_i - u)^2$$

Example 5.9 Solution

$E\left[-\frac{\partial^2 l}{\partial\mu^2}\right] = \frac{n}{\sigma^2}$

$E[Y_i] = \mu$ as $Y_i \sim N(\mu, \sigma^2)$

$E\left[-\frac{\partial^2 l}{\partial\mu\partial\sigma^2}\right] = E\left[\frac{1}{\sigma^4}\sum_{i=1}^{n}(Y_i-\mu)\right] = \frac{1}{\sigma^4}\sum_{i=1}^{n}\left[E(Y_i)-\mu\right] = \frac{1}{\sigma^4}\sum_{i=1}^{n}(\mu-\mu) = 0$

$$\begin{aligned}
E\left[-\frac{\partial^2 l}{\partial\sigma^4}\right] &= E\left[\frac{1}{\sigma^6}\sum_{i=1}^{n}(Y_i-\mu)^2\right] - \frac{n}{2\sigma^4} \\
&= \frac{1}{\sigma^6}E\left[\sigma^2\sum_{i=1}^{n}\left(\boxed{\frac{Y_i-\mu}{\sigma}}\right)^2\right] - \frac{n}{2\sigma^4} \\
&= \frac{1}{\sigma^4}E\left[\sum_{i=1}^{n}Z_i^2\right] - \frac{n}{2\sigma^4} \\
&= \frac{n}{\sigma^4} - \frac{n}{2\sigma^4} \\
&= \frac{n}{2\sigma^4}
\end{aligned}$$

$Z_i \sim N(0,1)$

So $\sum_{i=1}^{n} Z_i^2 \sim \chi_n^2$

$E\left[\sum_{i=1}^{n} Z_i^2\right] = n$

$$\therefore\ I_\theta = \begin{bmatrix} E\left[-\frac{\partial^2 l}{\partial\mu^2}\right] & E\left[-\frac{\partial^2 l}{\partial\mu\partial\sigma^2}\right] \\ E\left[-\frac{\partial^2 l}{\partial\sigma^2\partial\mu}\right] & E\left[-\frac{\partial^2 l}{\partial\sigma^4}\right] \end{bmatrix} = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}$$

Properties of the score and MLE

Theorem 13

Suppose $y_1, y_2, \ldots, y_n$ are independent observations with log-likelihood $\ell(\theta^*; \boldsymbol{y})$, where θ^* denotes the true value of the parameter. Under certain regularity conditions of $f(\boldsymbol{y}; \theta)$,

$= 0$

1. $E[S(\theta^*; \boldsymbol{Y})] = 0$
2. $\mathrm{var}\big(S(\theta^*; \boldsymbol{Y})\big) = I_{\theta^*}$. $= E\{[S(\theta^*; y)]^2\}$

   $$\frac{S(\theta; Y) - E(S(\theta; Y))}{\sqrt{Var(S(\theta; Y))}} \quad = I_\theta$$

3. The distribution of

$$\frac{S(\theta^*; \boldsymbol{Y})}{\sqrt{I_{\theta^*}}}$$

approximate,

converges to $N(0, 1)$ as $n \to \infty$.

$$E[S(\theta;y)] = E\left(\frac{\partial \log L}{\partial \theta}\right) = \int_{-\infty}^{+\infty} \frac{\partial \log L}{\partial \theta} \cdot L \, dy$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\cancel{L}} \frac{\partial L}{\partial \theta} \cdot \cancel{L} \, dy$$

under regularity condition

$$= \int_{-\infty}^{+\infty} \frac{\partial L}{\partial \theta} \, dy$$

$$= \frac{\partial}{\partial \theta} \underbrace{\int_{-\infty}^{+\infty} L \, dy}_{=1}$$

$$= \frac{\partial 1}{\partial \theta}$$

$$= 0$$

Theorem 14

Suppose $y_1, y_2, \ldots, y_n$ are independent observations with log-likelihood $\ell(\theta^*; \boldsymbol{y})$, where θ^* denotes the true value of the parameter. Under certain regularity conditions of $f(\boldsymbol{y}; \theta)$, then asymptotically

$$(\hat{\theta}_{MLE} - \theta^*)\sqrt{I_{\theta^*}} \to N(0,1)$$

$$\hat{\theta}_{MLE} \sim N(\theta^*, I_{\theta^*}^{-1}),$$

where $\hat{\theta}$ is the MLE for θ.

① $\hat{\theta}$ is an (asymptotically) unbiased estimator of θ

② An 'approximate' standard error for $\hat{\theta}$ is $\frac{1}{\sqrt{I_{\theta^*}}}$

BLUE.

③ $\hat{\theta}$ is (asymptotically) a minimum variance unbiased estimator of θ.

(because it achieves the Cramér-Rao lower bound and is unbiased asymptotically.)

In practice, θ^* is unknown. We approximate it with $\hat{\theta}$.

So an approximation of $\frac{1}{\sqrt{I_{\theta^*}}}$ is $\frac{1}{\sqrt{I_{\hat{\theta}}}}$.

Example 5.10

Suppose $y_1, y_2, \ldots, y_n$ are $i.i.d.\,Po(\lambda)$ observations.
Recall that $\hat{\lambda}_{MLE} = \bar{y}$ and $I_\lambda = \frac{n}{\lambda}$.
Theorem 14 states that

1. $\hat{\lambda}$ is asymptotically unbiased
2. The large-sample ~~standard error~~ is $\sqrt{\lambda/n}$
3. The distribution of

$$\frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}}$$

converges to $N(0, 1)$ as $n \to \infty$.

Check directly that the above is true.

Example 5.10 Solution

1. $E(\hat{\lambda}_{MLE}) = E(\bar{Y}) = \lambda$

2. $Var(\hat{\lambda}_{MLE}) = Var(\bar{Y}) = \frac{Var(Y_i)}{n} = \frac{\lambda}{n}$

3. CLT

$$\hat{\lambda}_{MLE} = \bar{Y} \rightarrow N\left(\lambda, \frac{\lambda}{n}\right)$$

So $$\frac{\hat{\lambda} - \lambda}{\sqrt{\frac{\lambda}{n}}} \rightarrow N(0, 1)$$

Confidence Intervals

Approximate confidence intervals

Suppose $y_1, y_2, \ldots, y_n$ are independent observations with log-likelihood $\ell(\theta; \boldsymbol{y})$.
An approximate $100(1-\alpha)\%$ confidence interval for θ is given by

$$\left(\hat{\theta} - z_{\alpha/2}\sqrt{I_{\hat{\theta}}^{-1}}, \hat{\theta} + z_{\alpha/2}\sqrt{I_{\hat{\theta}}^{-1}}\right).$$

$$\hat{\theta} \to N(\theta, I_{\theta}^{-1})$$

$$Z = \frac{\hat{\theta} - \theta}{\sqrt{I_{\theta}^{-1}}} \to N(0, 1)$$

The hypothesis test constructed using $\hat{\theta}$ above as the pivotal quantity is called the Wald test. The above CI is called as Wald confidence interval.

Wald test statistic

Suppose $Y_1, Y_2, \ldots, Y_n$ are i.i.d. observations with log-likelihood $\ell(\theta; \boldsymbol{Y})$.

The Wald test statistic for

$$H_0: \theta = \theta_0$$

is given by

$$Z = \frac{\hat{\theta} - \theta_0}{\sqrt{I_{\hat{\theta}}^{-1}}}. \xrightarrow{\text{under}} N(0,1)$$

If $H_0: \theta = \theta_0$ is true, then the distribution of Z converges to $N(0, 1)$ as $n \to \infty$.

A test with significance level approximately α is given by the rule:

Reject H_0 if $|Z| \geq z_{\alpha/2}$.

Example 5.11

Suppose $y_1, y_2, \dots, y_n$ are $i.i.d. Po(\lambda)$ observations.
Suppose we wish to test $H_0: \lambda = \lambda_0$.
a)Calculate the Wald test statistics.
b)Give an approximate $100(1-\alpha)\%$ confidence interval for λ.

Recall $\hat{\lambda}_{MLE} = \bar{y}$, $S(\lambda;\theta) = -n + \frac{1}{\lambda}\left(\sum_{i=1}^{n} y_i\right)$, $I_\lambda = \frac{n}{\lambda}$

① $$Z = \frac{\hat{\lambda} - \lambda_0}{\sqrt{I_{\hat{\lambda}}^{-1}}} = \frac{\bar{y} - \lambda_0}{\sqrt{\frac{\hat{\lambda}}{n}}} = \frac{\bar{y} - \lambda_0}{\sqrt{\frac{\bar{y}}{n}}}$$

$$\sqrt{I_{\hat{\lambda}}^{-1}}$$

② $$CI: \hat{\lambda} \pm Z_{\frac{\alpha}{2}} \sqrt{I_{\hat{\lambda}}^{-1}}$$

$$\bar{y} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\bar{y}}{n}}$$